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# Differential impact of disrupted child care on low-income US households during the COVID-19 pandemic, April 2021-February 2022

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## Abstract

**Introduction** The United States has a largely privatized child care system that stands out among high-income countries which generally incorporate some level of public support for child care. During the COVID-19 pandemic, the U.S. child care system was faced with unprecedented challenges which forced many child care providers to limit enrollment or cease operations. As child care providers re-opened, U.S. families faced a reality in which many child care providers reduced enrollment, some child care providers closed permanently, and nearly all child care providers faced multiple quarantines of staff and students. This left many families struggling to establish reliable child care.

**Research Aim** This study aimed to examine the impact of disrupted child care during the COVID-19 pandemic on U.S. households, with a particular focus on income disparities and related employment outcomes.

**Research Methods** This study analyzed nationally representative data from the U.S. Census Bureau's Household Pulse Survey (April 2021–February 2022) using descriptive and regression analyses to estimate the prevalence of child care disruptions and their association with household economic characteristics and employment impacts.

**Findings and Implications** In the 4 weeks preceding the survey between April 2021 and February 2022, we found caring for children was the primary reason survey respondents were not working. These disruptions had disparate impacts on households at or below 200% of the poverty threshold. At or below 200% of the poverty threshold households were more likely to report impacts of child care disruptions such as the loss of a job or the use of unpaid leave. This has left households at or close to poverty in increasingly dire economic circumstances and threatens the broader economic recovery of lower-income Americans. Findings underscore the importance of accessible and reliable child care for national economic security.

**Keywords** Child Care, COVID-19, Early childhood, Low-income households, National survey data

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## Introduction

During the COVID-19 pandemic, child care providers in the United States closed or operated at significantly reduced enrollment. Many households were abruptly left without child care and forced to balance employment and the supervision of children. How did child care disruptions during the COVID-19 pandemic influence parental employment and did disruptions differ by parental demographics and household characteristics? This paper uses national household data to examine the impacts of disrupted child care on economic factors such as employment and working hours in the United States. The economic implications of disrupted child care were particularly pronounced for low-income families, who were both the most likely to experience child care disruptions and the most vulnerable to associated employment losses, creating a compounding effect that exacerbated existing financial instability and workforce participation disparities. Understanding which populations were most impacted by child care disruptions is critical to targeting remediation supports for low-income households and to informing strategies that strengthen the U.S. child care system, recognizing child care as an essential service that underpins both family economic security and broader economic stability.

## Background on the U.S. child care system

The child care system in the United States is unique among high-income countries. While the majority of high-income countries provide some level of subsidy for child care, the United States has a largely private child care system in which private individuals or companies provide child care under varying levels of regulation and oversight. In general, much of the regulation of the U.S. child care system lies at the State level, with each State establishing their own system to license independent child care providers. Many child care providers operate larger center-based child care, however, an increasing number of children are enrolled in home-based child care. In 2019, 21% of infants and toddlers were cared for in a child care center while 29.4% were in home-based child care, the remaining 49.6% were in the care of their parents (Li & Barnett, 2021). In the U.S., home-based child care is a collection of predominantly unregulated child care in individual homes. In 2019, roughly 5.2 million child care providers operated in a home-based child care setting. Of these 5.2 million providers, only 91,200 were listed with the State and subject to regulations. The remaining 5.08 million providers were unlisted child care providers that are not registered with the State and not regulated (Datta, Milesi, & Zapata-Gietl, 2021). This patchwork of private providers operating under varying regulations creates a complex system that is particularly challenging to support during a National-level emergency such as the COVID-19 pandemic. At a national level, deciphering child care closures during the COVID-19 pandemic among unlisted child care providers is nearly impossible. Many likely chose to follow government recommendations and close for some period of time, but little can be known about the duration or extent of child care closures among unlisted child care providers at the national level.

At the Federal level, two major programs exist to support child care for low-income families: the Child Care and Development Fund (CCDF), which provides funding to states, territories, and Tribes to help low-income families afford child care, and Head Start, which provides grants to child care providers to serve low-income families and families with foster children. In 2019, prior to COVID-19 pandemic supplements, States

and Territories spent a total of $10.3 billion through CCDF to assist families in affording child care (Office of Child Care, 2021). Although CCDF is one of the Nation's largest child care subsidies, the program has struggled to reach all eligible families. Of the 13.5 million children eligible for CCDF support, only 1.9 million (14%) actually received the funds (Chien, 2020). The Head Start program serves roughly 1 million low-income and foster children every year and is available to all children with family incomes at or below the poverty threshold, foster children regardless of income, homeless children, and children from families receiving certain Federal social service benefits (Office of Head Start, 2020; Office of Early Child Development, 2020). Head Start providers are generally subject to both Federal and State oversight, although some States do not provide oversight for Head Starts and accept Federal oversight as sufficient (Bernstein et al., 2022).

For both private child care providers and Federally subsidized child care providers, wages for child care workers are low. Despite the high cost of child care in the United States, the median salary for a child care provider in 2021 was $27,490 per year (Bureau of Labor Statistics, 2022a). For comparison, the median salary for Kindergarten and Elementary school teachers in 2021 was $61,360 per year (Bureau of Labor Statistics, 2022b). More than 15% of child care workers live below the poverty line and half of child care workers rely on public assistance (Ho & Boak, 2021) despite the national mean child care cost of $10,174 per child per year in 2020 (Childcare Aware of America, 2022a).

### Economics of child care

Child care in the United States is a major expense for many families with children. The high cost of child care may have a detrimental effect on the supply of parents in the workforce, particularly for women (Gurrentz, 2021; Olivetti & Petrongolo, 2017). Reasons ranging from gaps in wages earned between men and women to parenting expectations often push female caregivers to leave the workforce - notably, when families are struggling to pay child care costs. Other high-income countries have historically sought to increase female workforce participation through child care subsidies (Barigozzi et al., 2017). Among OECD countries, the average investment in child care subsidies is around 0.7% of GDP whereas the U.S. spends around 0.2% of GDP on subsidizing child care. This equates to government subsidies at around $200 per year for most families with toddlers, typical in the form of an annual tax credit. Conversely, among OECD countries with available data, child care subsidies range from $3,327 per year in Israel to $29,726 per year in Norway (Miller, 2021). An example of a child care subsidy program that has shown success is in the Canadian territory of Quebec. Quebec passed a substantial child care subsidy program in 1997 which provided child care for all 4-year old children for $5.00 per child per day, and extended free kindergarten from half-day to full-day. The result was an increase in women with at least one child under age 5 in the workforce by 8% points in 2002 (Lefebvre & Merrigan, 2008).

### Impacts of COVID-19 on the U.S. child care system

The child care system in the United States was largely considered to be in crisis as a result of soaring costs and inadequate supply long before the COVID-19 pandemic began in early 2020. The COVID-19 pandemic presented an unprecedented challenge to the child care industry. At the national level, the U.S. Centers for Disease Control and Prevention

released multiple iterations of COVID-19 prevention and mitigation guidance for child care operators which included more stringent cleaning protocols, increased ventilation, decreased enrollment, and quarantine procedures (CDC, 2022). At the state level, many states chose to close the majority of early childhood care and education facilities for undetermined amounts of time. States that did close child care often set up a handful of child care facilities for essential workers, where individuals deemed to be "essential" such as those in the medical or law enforcement profession, could take their children for care. Other states chose to enforce strict protocols and enrollment limitations, limiting the number of children that could be in a child care facility at any given time (Council For Strong America, 2020; Childcare Aware of America, 2022b).

During the COVID-19 pandemic, an unprecedented number of child care providers closed temporarily or permanently. As of 2020, 53% of States reported a decline in center based child care providers and 84% of States reported a decline in home-based child care providers. Between 2019 and 2021, there was an estimated overall 9% decrease in center-based child care centers across the country and an overall 10% decrease in home-based child care across the country (Childcare Aware of America, 2022a). One analysis examining closures of child care facilities during the COVID-19 pandemic found that at a peak, 70% of U.S. child care facilities were closed or operating at less than 50% capacity in April 2020. As of April 2021, that rate was halved to 35% of child care facilities in the U.S. closed or operating at 50% capacity (Lee & Parolin, 2021). This loss in child care represents a significant challenge for families and created challenges in maintaining employment for caregivers.

Disruptions to child care during the COVID-19 pandemic also had considerable impacts on the U.S. economy, and clearly demonstrated the value of child care in supporting State and National economies. In the State of Texas alone, disruptions to child care during the COVID-19 pandemic which forced parents to miss work, drop out of work, and miss educational opportunities are estimated to cost the State economy $9.39 billion annually. This breaks down into a $1.8 billion dollar loss in tax revenue for the State of Texas, and a $7.59 billion loss in employee absences and employee turnover across the State (U.S. Chamber of Commerce Foundation, 2021).

#### Importance of child care for children, parental employment, and the economy

Aligned with Becker's (1964) seminal work on human capital theory, affordable and reliable child care provides an essential avenue for parents to increase their capital through education, training, and work experience. In turn, this provides support to the broader economy through increased incomes, increased tax revenue, increased consumer spending and decreased reliance on the social safety net. The value of affordable reliable child care goes beyond theoretical and has been demonstrated through several national examinations of the impacts of child care on the national economy. Expanding access to affordable child care produces strong returns by increasing labor supply, boosting tax revenues, and enhancing overall productivity. When child care costs are reduced and availability improves, more parents enter or remain in the workforce, leading to higher earnings and greater economic output at the macro level (Blau & Currie, 2006; Bauernschuster & Schlotter, 2015; Herbst, 2010).

The child care industry supports the consumption of goods, which is of particular importance in a consumer driven economy such as the United States. The increased

consumption of goods is driven by increased incomes among families with children in care, increased consumption by employed workers in the child care industry, and increased consumption by the school itself. In one small region of 7 counties in Northern Minnesota, analysts estimated the direct economic impact of all goods and services consumed by the child care industry to total $30 million per year (Chase et al., 2018). Nationally, the child care industry generates an estimated $47.2 billion in revenue and employs 1.52 million workers (Committee for Economic Development, 2019).

Beyond direct and immediate implications for the U.S. economy, child care has long term impacts on the economy. Quality early childhood education and care produces an estimated 13% return on investment (Garca et al., 2020). Returns on investment are often seen as children age. Children that attend high quality early childhood care and education are likely to enjoy better health and overall quality of life, have increased incomes and subsequently produce more tax revenue, less likely to engage in crime, more likely to complete their education, and mothers of children in child care are more likely to engage in the workforce (Garca et al., 2020).

This analysis seeks to examine some of the impacts of disrupted child care for families when child care facilities closed or operated with decreased enrollment using the first available nationally representative data. There is evidence that the loss of child care between April 2021 to February 2022 during the COVID-19 pandemic had significant consequences on households with children. One analysis showed American households with children lost an estimated $13 billion in income for each year of the COVID-19 pandemic (Belfield & Kashen, 2022). This analysis seeks to understand if economic impacts of the child care disruptions were felt equally among households, or if some households struggled more than others. Given the economic implications of child care, this examination speaks to the cost-benefits of child care subsidies and how to target subsidies to maximize impact.

## Methods

Data came from the U.S. Census Bureau's Household Pulse Survey (HPS) Phases 3.1 through 3.3 when child care questions were added to the HPS beginning in April 2021. The HPS launched in April 2020 to better understand the impact of the COVID-19 pandemic on U.S. households. Households were randomly sampled using the Census Bureau's Master Address File, and adult respondents were invited by text and/or email to participate in a web-based survey. Data were collected on a rolling basis in survey periods of 12 days with data publicly released every two (Phases 3.1 and 3.2) or four (Phase 3.3) weeks to provide near real-time information. Phase 3.1 began April 14, 2021, and Phase 3.3 ended Feb 7, 2022. Data were not collected between phase implementation (Phase 3.1 to 3.2: Jul 6–20, 2021; Phase 3.2 to 3.3: Oct 12 – Nov 30, 2021). During each survey period, between 57,000 and 83,000 adults responded and about one-third of adult respondents resided in households with children (U.S. Census Bureau, 2022). Data are publicly available through the U.S. Census Bureau.

During Phase 3.1 (Apr 14 – Jul 5, 2021), respondents were asked "At any time in the last 4 weeks, were any children in the household unable to attend day care or another child care arrangement because of the coronavirus pandemic?". During Phases 3.2 and 3.3 (Jul 21, 2021 – Feb 7, 2022), respondents were asked "At any time in the last 4 weeks, were any children in the household unable to attend day care or another child care

arrangement as a result of child care being closed, unavailable, unaffordable, or because you are concerned about your child's safety in care?" (U.S. Census Bureau, 2022). Households were included if a respondent answered yes or no ($N$ = 16,804–23,640). Data from Phases 3.1 through 3.3 were combined, though the questions differed slightly, because patterns in estimates of disrupted child care persisted across covariates. Estimates for disrupted child care and study covariates were descriptively compared across phases, and z-tests were estimated for key indicators to assess significant differences prior to combining. Further, the constructs measured by the different questions were considered sufficiently similar and the increase in the proportion of disrupted child care was considered sufficiently small enough to combine the phases. Employment impacts included the primary reason the respondent wasn't working for pay or profit in the last 7 days and impact of disrupted child care on the respondent's job. Sociodemographic characteristics included race/Hispanic ethnicity, education, and marital status of the adult respondent; sociodemographics were not collected on children in the household. Household characteristics included household income as a percent of the poverty threshold (household income) and whether the household was caught up on rent or mortgage, difficulty paying usual household expenses, and whether any children were not eating enough because the household couldn't afford food. The HPS did not collect demographic characteristics of children in the household, such as child age or race/Hispanic ethnicity. Because child's age was not collected, any child that typically had a child care arrangement was included.

We estimated the proportion of households that reported any child was unable to attend day care or another child care arrangement or "child care disruptions", tested for significant associations between child care disruptions and sociodemographic and household characteristics, and described employment impacts of child care disruptions overall and stratified by race/Hispanic ethnicity for data collected from Dec 29, 2021 through Feb 7, 2022, the latest available data at the time of this analysis. Chi-square tests were conducted to assess statistical significance in descriptive analyses between each outcome of interest (i.e. child care disruptions and employment impacts) and study covariates. Additionally, we described trends in rolling averages of approximately one month of child care disruptions overall and stratified by household poverty status across phases; within each phase, two survey periods were combined per average to address small sample sizes. Likelihood ratio tests using logistic regression were conducted to assess statistical significance of trends. We report frequencies, proportions, 95% confidence intervals (CI), and p-values where appropriate for all analyses. All analyses used a two-sided significance level of 0.05. Estimates were weighted for nonresponse and to match the Census Bureau's estimates of the U.S. population by age, sex, race, Hispanic ethnicity, and education level. Compared to the U.S. population, respondents were more likely to be female, older, non-Hispanic white or non-Hispanic other race, and have more education – at least some college or bachelor's degree or higher (U.S. Census Bureau, 2022). Analyses were conducted in SAS version 9.4 using survey procedures and incorporating 80 replicate weights with Fay's adjustment to account for complex survey design.

## Results

### Overall child care disruptions reported from Dec 29, 2021 through Feb 7, 2022

Almost half of U.S. households with any children had at least one child in a day care or child care arrangement (46.9%, 95% CI: 46.1, 47.7; data not shown). Respondents that reported at least one child in a child care arrangement were primarily non-Hispanic White (54%, 95% CI: 53.4, 55.3) followed by Hispanic (19.1%, 95% CI: 18.4, 19.8) and non-Hispanic Black (16.8%, 95% CI: 15.8, 17.7), married (65.1%, 95% CI: 64.0, 66.2) (Table 1).

Overall, 32.4% (95% CI: 31.4, 33.3) of U.S. households with a child care arrangement experienced child care disruptions. We found significant differences in child care

**Table 1** Distribution of households with children in a child care arrangement and the prevalence of disrupted child care, overall and by sociodemographic and household characteristics, Census Household Pulse Survey 3.3 (Dec 29, 2021 – Feb 7, 2022)

| | Study sample | | Disrupted child care among households with child care | | |
|---|---|---|---|---|---|
| | No. | % (95% CI)[1] | No. | % (95% CI)[1] | *p* |
| Overall | 23,640 | 100.0 | 7,739 | 32.4 (31.4, 33.3) | |
| Adult respondent characteristics | | | | | |
| Race/Hispanic Ethnicity | | | | | < 0.01 |
| Asian/Pacific Islander, NH | 1638 | 5.9 (5.6, 6.2) | 462 | 27.0 (24.2, 29.8) | |
| Black, NH | 2272 | 16.8 (15.8, 17.7) | 799 | 36.0 (33.2, 38.9) | |
| Hispanic | 3167 | 19.1 (18.4, 19.8) | 1050 | 32.4 (29.5, 35.4) | |
| White, NH | 15,439 | 54.3 (53.4, 55.3) | 5025 | 31.5 (30.3, 32.8) | |
| Multi-race/Other, NH | 1124 | 3.9 (3.5, 4.2) | 403 | 36.0 (32.1, 39.9) | |
| Education | | | | | < 0.01 |
| ≤High school degree/GED | 3260 | 35.9 (34.9, 36.9) | 858 | 27.4 (25.3, 29.6) | |
| Some college/Associate's degree | 6911 | 30.1 (29.2, 30.9) | 2273 | 34.5 (33.0, 36.0) | |
| Bachelor's degree or higher | 13,469 | 34.0 (33.3, 34.7) | 4608 | 35.6 (34.3, 36.9) | |
| Marital Status | | | | | < 0.01 |
| Married | 17,130 | 65.1 (64.0, 66.2) | 5458 | 30.7 (29.7, 31.6) | |
| Not Married | 6431 | 34.9 (33.8, 36.0) | 2259 | 35.7 (33.8, 37.6) | |
| Health Insurance | | | | | 0.25 |
| Private insurance | 15,332 | 61.6 (60.3, 62.9) | 4997 | 32.4 (31.2, 33.6) | |
| Any public insurance | 4950 | 28.8 (27.8, 29.8) | 1710 | 34.0 (31.8, 36.3) | |
| No insurance | 1157 | 9.6 (8.7, 10.5) | 347 | 29.7 (24.3, 35.0) | |
| Households characteristics | | | | | |
| Household income | | | | | 0.37 |
| ≤ 138% | 3360 | 26.4 (25.3, 27.6) | 1178 | 34.3 (31.1, 37.5) | |
| > 138% – 200% | 2412 | 15.3 (14.2, 16.3) | 772 | 33.4 (29.8, 37.0) | |
| > 200% – 400% | 4989 | 23.8 (22.7, 24.9) | 1558 | 31.1 (29.4, 32.8) | |
| > 400% | 9445 | 34.5 (33.5, 35.5) | 3186 | 33.2 (31.5, 34.9) | |
| Caught Up on Rent/Mortgage | | | | | < 0.01 |
| Yes | 18,661 | 84.4 (83.6, 85.2) | 5910 | 31.2 (30.1, 32.3) | |
| No | 2198 | 15.6 (14.8, 16.4) | 970 | 41.5 (37.9, 45.1) | |
| Difficulty Paying Usual Household Expenses (last 7 days) | | | | | < 0.01 |
| Very/Somewhat difficult | 7158 | 40.3 (39, 41.5) | 3121 | 42.5 (40.6, 44.5) | |
| Not at all/a little difficult | 15,737 | 59.7 (58.5, 61) | 4388 | 25.5 (24.5, 26.6) | |
| Children Not Eating Enough Due to Lack of Affordability (last 7 days) | | | | | < 0.01 |
| Often/Sometimes true | 2076 | 12.8 (12, 13.6) | 964 | 44.9 (41.2, 48.7) | |
| Never true | 20,239 | 87.2 (86.4, 88) | 6354 | 30.7 (29.7, 31.7) | |

[1]Frequencies are unweighted, percentages are weighted

disruptions by respondent race and Hispanic ethnicity, education, and marital status. The proportion of U.S. households experiencing child care disruptions was higher among respondents who were non-Hispanic Black (36.0, 95% CI: 33.2, 38.9) and non-Hispanic other (36.0, 95% CI: 32.1, 39.9), had a bachelor's degree or higher (35.6%, 95%CI: 34.3, 36.9), and were not married (35.7%, 95% CI: 33.8, 37.6). Child care disruptions were least frequently reported among respondents who were non-Hispanic Asian/Pacific Islander (27.0%, 95% CI: 24.2, 29.8), had a high school degree or less education (27.4, 95% CI: 25.3, 29.6), and were married (30.7, 95% CI: 29.7, 31.6).

The prevalence of child care disruption was similar across household income. In contrast, child care disruption was significantly higher among respondents who were not caught up on rent/mortgage (41.5%, 95% CI: 37.9, 45.1), had difficulty paying usual household expenses (42.5%, 95% CI: 40.6, 44.5), and who reported children were not eating enough in the last 7 days because they couldn't afford enough food (44.9, 95% CI: 41.2, 48.7).

The most common impacts of child care disruptions on employment were the respondent used paid leave (35.5, 95% CI: 33.5, 37.5), cut hours (34.9, 95% CI: 33.1, 36.7), took unpaid leave (28.6, 95% CI: 26.8, 30.4), and supervised children while working (28.6, 95% CI: 26.7, 30.5; Table 2). However, these impacts significantly differed by whether households lived above or below 200% of the poverty threshold. Households living at or below 200% of the poverty threshold were more likely than households living above 200% of the poverty threshold to take unpaid leave (31.3 vs. 26.3%; $p = 0.02$) to leave their job (17.6% vs. 8.9%; $p < 0.001$) and to lose their job (9.7% vs. 1.6%; $p < 0.001$). In contrast, households living above 200% of the poverty threshold were more likely to use paid leave (48.3% vs. 22.5%; $p < 0.001$), supervise children while working (44.1% vs. 14.4%; $p < 0.001$), and cut hours (42.5% vs. 28.6%; $p < 0.001$). Further, respondents reported that caring for children not in school or day care was the primary reason for not working in the last 7 days (Fig. 1) with significant difference by race and Hispanic ethnicity (Fig. 2). Non-Hispanic White (26.6%, 95% CI: 24.5, 28.8) and non-Hispanic Asian/Pacific Islander (26.1%, 95% CI: 19.0, 33.3) respondents were the most likely, and non-Hispanic Black (17.7%, 95% CI: 14.8, 20.5) respondents were the least likely, to not work in the last 7 days to care for children not in school or day care.

**Table 2** Frequencies, proportions, and 95% CIs of employment impacts due to child care disruptions overall and by household income among households with children in a child care arrangement that experienced disrupted child care, Census Household Pulse Survey 3.3 (Dec 29, 2021 – Feb 7, 2022)[1]

| | **Study sample** | | **Household income** | | | | |
|---|---|---|---|---|---|---|---|
| | **Overall** | | **≤ 200% FPL** | | **> 200% FPL** | | |
| | No. | % (95% CI)[1] | No. | % (95% CI)[2] | No. | % (95% CI)[2] | p |
| Employment impacts | | | | | | | < 0.01 |
| Took unpaid leave | 2078 | 28.6 (26.8, 30.4) | 686 | 31.3 (27.7, 34.8) | 1088 | 26.3 (24.1, 28.4) | |
| Used paid Leave | 3210 | 35.5 (33.5, 37.5) | 490 | 22.5 (19.0, 26.0) | 2381 | 48.3 (46.0, 50.6) | |
| Cut hours | 2875 | 34.9 (33.1, 36.7) | 626 | 28.6 (25.6, 31.5) | 1946 | 42.5 (40.3, 44.7) | |
| Left job | 822 | 13.1 (11.8, 14.5) | 332 | 17.6 (14.9, 20.2) | 374 | 8.9 (7.6, 10.2) | |
| Lost job | 264 | 4.8 (3.9, 5.7) | 166 | 9.7 (7.4, 12.1) | 62 | 1.6 (0.9, 2.3) | |
| Did not look for a job | 959 | 12.8 (11.6, 14.1) | 359 | 16.3 (13.7, 19.0) | 491 | 10.3 (9.1, 11.5) | |
| Supervised children while working | 2887 | 28.6 (26.7, 30.5) | 382 | 14.4 (12.0, 16.7) | 2261 | 44.1 (41.3, 46.8) | |

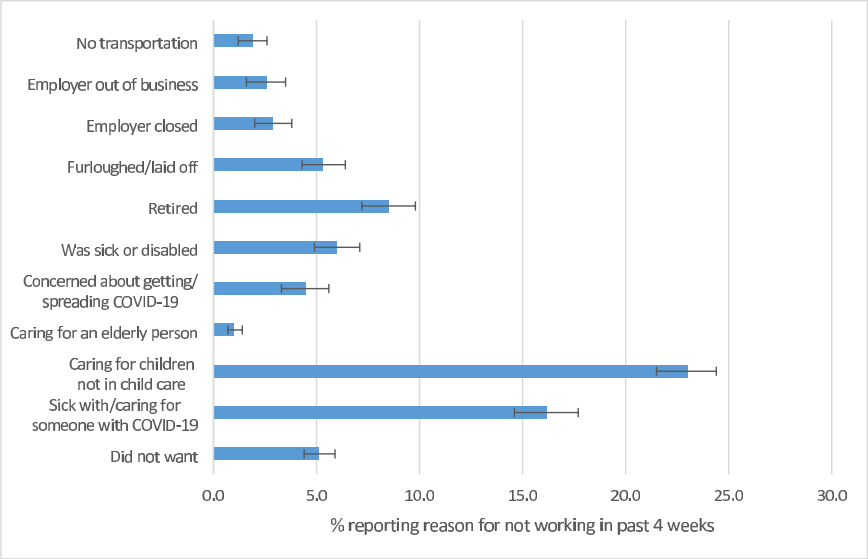



**Fig. 1** Primary reason for not working in the past four weeks among households with children in a child care arrangement, Household Pulse Survey (Dec 29, 2021 - Feb 7, 2022)

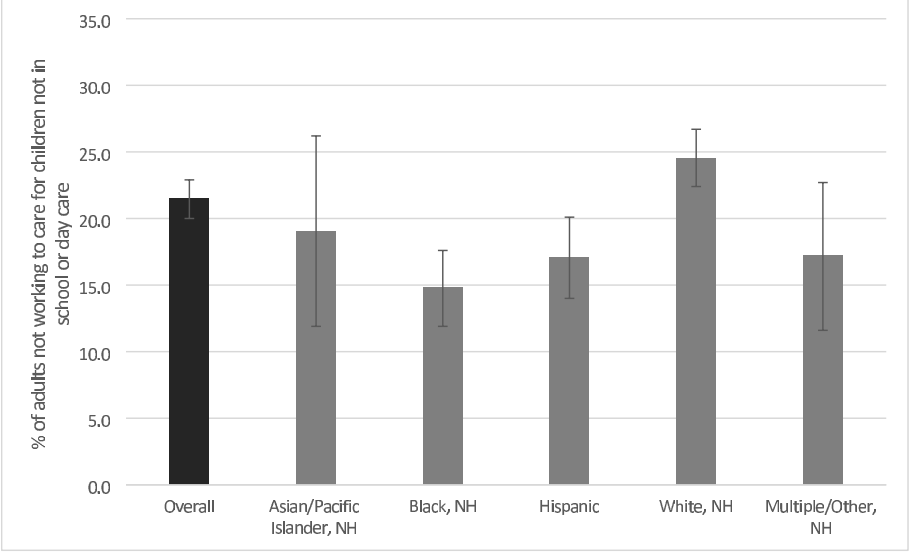



**Fig. 2** Primary reason for not working in the past four weeks was to care for children not in school or day care by Race/Hispanic Ethnicity, Household Pulse Survey (Dec 29, 2021 - Feb 7, 2022)

### Trends in child care disruptions from Apr 14, 2021 through Feb 7, 2022

From April 14 through May 10, 2021, about 1 in 5 U.S. households with a child care arrangement experienced child care disruptions (20.4%, 95% CI: 19.1, 21.7%; data not shown). The prevalence of child care disruptions decreased to 17.8% (95% CI: 16.7, 18.8) from June 7 through July 5, 2021, the lowest prevalence during Phase 3.1. When the updated survey item was fielded on July 21, 2021, the prevalence of child care disruptions increased to 25.3% (95% CI: 24.3, 26.4) and remained steady through October 11, 2021. In December 2021, child care disruptions increased and reached peak prevalence from January 26 through Feb 7, 2022 (34.8%, 95% CI: 33.4, 36.2). Trends in child care disruptions by race and Hispanic ethnicity mirrored overall trends from December 29,

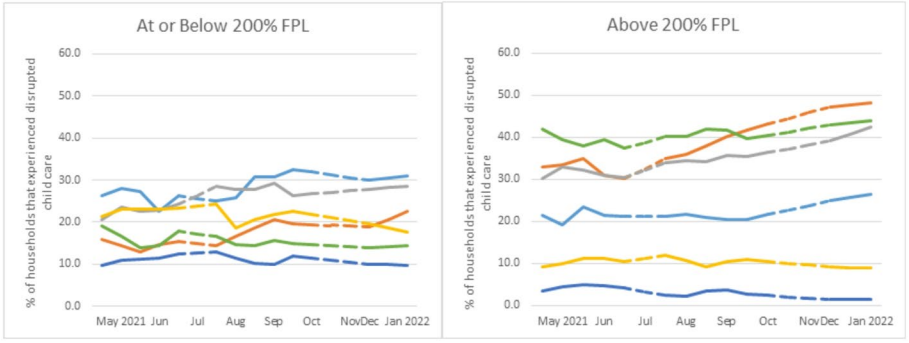



**Fig. 3** Trends in employment impacts due to disrupted child care by poverty threshold among households that experienced disrupted child care, Household Pulse Survey (Apr 14, 2021 - Feb 7, 2022)[1,2] [1]Dashed lines reflect periods between phases when data were not collected. [2]Question identifying study sample changed from Phase 3.1 "At any time in the last 4 weeks, were any children in the household unable to attend day care or another child care arrangement because of the coronavirus pandemic?" to Phases 3.2 and 3.3 "At any time in the last 4 weeks, were any children in the household unable to attend day care or another child care arrangement as a result of child care being closed, unavailable, unaffordable, or because you are concerned about your child's safety in care?"; denominator is the respondents who answered "yes"

2021 through February 7, 2022. Non-Hispanic White respondents were consistently among the least likely, and non-Hispanic Black respondents were the most likely, to experience disruptions in child care. Child care disruptions varied over time for non-Hispanic Asian/Pacific Islander respondents with some periods of disruptions as high as non-Hispanic Black respondents (Apr 14 – Jul 5, 2021), some periods almost as low as non-Hispanic White respondents (Jul 21 – Oct 11, 2021), and some periods less than non-Hispanic White respondents (Dec 1, 2021 – Feb 7, 2022).

Similar to trends in child care disruptions by race and Hispanic ethnicity, trends by household income mirrored overall estimates from Dec 29, 2021 through Feb 7, 2022. Households living at or below 200% of the poverty threshold, were significantly more likely to report child care disruptions than households living above 200% of the poverty threshold from April 14, 2021 through January 10, 2022. The prevalence of child care disruption was similar among households at or below (34.0%, 95% CI: 31.7, 36.2) vs. above 200% of the poverty threshold (32.3%, 95% CI: 31.2, 33.5) from December 29, 2021 through February 7, 2022. Patterns of the impact on employment due to child care disruptions by household income described from Dec 29, 2021 through Feb 7, 2022 were also similar from April 14, 2021 through February 7, 2022 (Fig. 3). Households at or below 200% of the poverty threshold were more likely than households above 200% of the poverty threshold to take unpaid leave, leave their job, or lose their job as a consequence of child care disruptions. Households above 200% of the poverty threshold were more likely to supervise children while working followed by use paid leave and cut hours from April 14 through September 27, 2021.

## Discussion

During the height of the Omicron variant COVID-19 surge, December 2021 through February 2022, almost half of households had at least one child in a child care arrangement. Almost one-third of these families experienced child care disruptions during this time period, which had a considerable impact on caregiver employment. The primary reason for not working among households with children was to care for children not in school or child care. Additionally, these disruptions were more prevalent

for non-married households, Black households, and households that were experiencing economic hardships (i.e., those behind on rent/mortgage, had difficulty paying for usual household expenses, and those experiencing food scarcity). Households living at or below 200% of the poverty threshold were more likely to take unpaid leave, leave their job, or lose their job, while households above 200% of the poverty threshold were more likely to use paid leave and supervise their children while working. These outcomes align with the working conditions of many low-income jobs and the benefits that are often available to low-income workers. Low-income jobs are often inflexible and rarely provide benefits like paid leave or telework. Higher income jobs often come hand-in-hand with greater benefits, providing families the option to take paid leave and telework, or adjust their hours to take care of children when disruptions occur (Bureau of Labor Statistics, 2021). Findings presented here support findings throughout the COVID-19 pandemic which highlighted, at various timepoints, a significant percentage of families were severely impacted by disrupted child care and the burden of disrupted child care was most acute in low-income households and households experiencing material hardship (Sappenfield et al., 2023; Padilla & Lin, 2024). Recent findings also point toward variations in disrupted child care may exist across states, with notably high levels of disruptions in New Mexico, Washington, D.C., California, and Alaska (Batioja et al., 2023). Strikingly, disruptions in child care during the COVID-19 pandemic have been linked to lower literacy scores by age 6 and higher levels of internalizing behaviors (Miller et al., 2025). Global research has highlighted the importance of these early warning signs in child behavior, as child care instability during the COVID-19 pandemic has been linked to children's stress and internalizing behaviors, showing signs of an indirect association with less closeness in parent-child relationships 3-years later (Moed & Razer, 2025).

Low-income families face disproportionate risks when child care systems falter, in part because they are less likely to have access to benefits such as paid leave or telework and often work in jobs with rigid schedules (Bureau of Labor Statistics, 2021 b). As a result, child care disruptions jeopardize employment and put families at higher risk of material hardships, including food insecurity and housing instability (Belfield & Kashen, 2022). The employment consequences of child care disruptions—such as job loss, reduced hours, and unpaid leave—compound existing inequalities through lost wages for low-income families who are less likely to have sufficient savings to cover the lost wages (Child Care Aware of America, 2022a). Because low-wage workers are heavily represented in essential service industries, instability in their access to child care not only affects individual families but also undermines community and national economic resilience (U.S. Chamber of Commerce Foundation, 2021). Parental health and wellbeing also suffer during periods of disrupted child care, particularly for female parents. During the COVID-19 pandemic, households with disruptions to child care saw female caregivers reporting decreased levels of mental and physical wellbeing (MacCallum-Bridges et al., 2024). These findings underscore that child care is not merely a family responsibility but critical social infrastructure, representing an essential tool for national economic security (Garcia, Heckman, Leaf, & Prados, 2020). The disproportionate effects of instability in the U.S. child care system on low-income households are not unique to the COVID-19 pandemic—similar patterns can be expected during other crises, including natural disasters, seasonal disease outbreaks, or economic downturns—unless stronger

safety nets and structural supports are implemented (Child Care Aware of America, 2022a).

These data demonstrate the impact that the COVID-19 pandemic had on U.S. households with children and speak to the child care industry's ability to withstand such crises. Although global pandemics, such as the COVID-19 pandemic, represent a rare event; catastrophic and unexpected crises are unfortunately less rare. Natural disasters, virus outbreaks such as influenza or RSV, and economic crises are a reality that many communities will face each year. Findings presented here highlight the fragility of the U.S. child care industry, particularly for the most vulnerable families.

#### Limitation and implications

This study had multiple limitations. This analysis relied on secondary data collected roughly one year into the COVID-19 pandemic. It is likely that data would have demonstrated different impacts had these data been collected in the first year of the COVID-19 pandemic. Furthermore, data is limited to households with an address and collected online. This limits data collected to only those individuals and families with a home address and internet access, omitting some individuals and families in extreme poverty. However, recent data suggests 96–99% of adults aged 18–64 have access to internet and report using internet in 2021 as a result of an increase in smartphone access over the last decade (Vogels 2021a, b). The HPS did not collect information on children's characteristics such as child's age, which limit interpretation of results and generalizability. For example, households with young children who are in full day child care are likely combined with households with older children in afterschool programs. Disruptions to either care arrangement could have different impacts on households. While access to child care can vary by children's characteristics (e.g., infant care is more expensive and harder to find than preschool or afterschool care), household information such as income, as is used as one of this study's indicators, are also associated with access to child care. Lastly, to address differential participation among respondents, data were weighted to the U.S. population by demographics, which should mitigate nonresponse bias (Peterson et al., 2021).

Despite limitation, these data demonstrate the significant impact that the COVID-19 pandemic had on U.S. households with children. In March 2021, the American Rescue Plan Act included an additional $39 billion in federal funding for child care in order to help address the findings identified retrospectively in this analysis. However, these data indicate that his infusion of federal support might not have been enough. The finding that households living at or below 200% of the poverty threshold were more likely to take unpaid leave, leave their job, or lose their job suggests system-level paid leave policies are needed to buffer low-income families from child care disruptions. The FAMILY Act is one such system-level policy approach that would create a permanent, national paid family and medical leave program (Act, 2025). Illinois' Paid Leave for All Workers Act is another system-level policy approach that allows most workers to accrue up to 40 h of paid leave to use for any reason (Illinois Department of Labor, 2025). The finding that disruptions were more prevalent for households that were experiencing economic hardships (i.e., those behind on rent/mortgage, had difficulty paying for usual household expenses, and those experiencing food scarcity) also suggests that updated system level tax policies and subsidy reforms are needed to support families during times of crisis.

The Child Tax Credit currently does not reach families with incomes too low to report and the Child and Dependent Care Tax Credit benefit levels have not been updated since 2001, failing to keep pace with inflation and the price of child care (Boyle et al., 2021). Low-income families would also benefit from expansions and changes to subsidy policies and programs. One common subsidy reform often proposed is universal child care and/or preschool which guarantees child care for all children and removes market factors to ensure that all families have access to care. These system-level and subsidy reforms to new and existing programs could support families in times of crisis such as was experienced during the COVID-19 pandemic.

Additionally, these data speak to the child care industry's ability to withstand crises, such as the COVID-19 pandemic. Even with an infusion of federal support including $24 billion in Child Care Stabilization Grants for child care providers to reopen or stay open child care providers still closed or were unavailable for families (NAEYC, 2022). These efforts might not have been enough. Recent workforce stabilization policy reforms aim to buffer the child care industry from such crises. The 2024 Head Start final rule updated Performance Standards to require programs to provide competitive wages for staff and provide or facilitate access to affordable health insurance and offer paid leave (Supporting the Head Start Workforce and Consistent Quality Programming, 2024). The 2024 Child Care final rule improved payment practices for child care providers to help stabilize operations by requiring payment based on enrollment and building supply via grants and contracts. Findings from the current study indicating that child care disruptions remained prevalent through February 2022 demonstrate the fragility of the child care market and why these child care workforce stabilization and subsidy reforms might be necessary.

This analysis demonstrates that many low-wage workers suffered disproportionately severe impacts from disrupted child care, and the uniquely private child care system in the United States is highly vulnerable during unforeseen events and crises. Although the COVID-19 pandemic represents an extreme event, similar outcomes can be expected during natural disasters, severe cold and flu outbreaks, and other semi-regular crises. Data demonstrated the impact of the disruption and loss of child care on families, and indicates child care is essential to maintain uninterrupted employment for caregiver which has impacts for the broader economy.

**Author Contribution**

Authors' Contributions: The authors confirm contribution to this paper as follows: Anne Day Leong designed the study, conducted the literature review, and wrote the background section. Olivia Sappenfield conducted data analysis and wrote the methods and results sections. All authors contributed to the discussion, conclusion, and limitations sections. Francesca Longo provided expertise on the child care industry and wrote the discussion section.

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**Data Availability**

Data used for this publication are publicly available at the U.S. Census Bureau website at https://www.census.gov/data/experimental-data-products/household-pulse-survey.html

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